

PRADEEP GANDU

Cloud Data Engineer · Automotive Software & AI/ML

Chemnitz, Germany (Open to Relocation) · German — C1 | English — Fluent · +49 176 59677415 · gandupradeep2026@gmail.com · [LinkedIn](#) · [GitHub](#)

· Portfolio

Master's student in Automotive Software Engineering at TU Chemnitz (90/120 ECTS completed) with an engineering focus on cloud data engineering (GCP/AWS), distributed stream processing (Apache Beam, PySpark), and autonomous vehicle systems (CARLA, CAN-Bus UDS). Proven track record developing production-style streaming pipelines, explainable vehicle diagnostics audited on 39k+ real OBD records, and automated CI/CD workflows. Certified German C1 & English Fluent. Eligible for Master Thesis, Working Student, and Full-Time permanent roles in Germany.

TECHNICAL SKILLS

Cloud & Big Data:	GCP (BigQuery, Pub/Sub, Dataflow, Dataproc, Cloud Storage, IAM), AWS (EC2, S3, RDS), Apache Beam, PySpark, Spark SQL, Apache Airflow, HDFS, Parquet.
Software & Tools:	Python, SQL, FastAPI, Docker, Docker Compose, Pydantic, Streamlit, Linux/Bash, Git/GitHub Actions CI/CD, Pytest, Ruff, MyPy.
Automotive & AI:	CARLA Simulator, 3D Trajectories, CAN-Bus, UDS (ISO 14229), DBC Networks, SocketCAN, IsolationForest, RAG, Ollama (Qwen3), MCP, YOLOv8.

SELECTED TECHNICAL PROJECTS

Automotive Telemetry Streaming Platform (GCP) [GCP Pub/Sub, Apache Beam / Dataflow, BigQuery, Airflow, Pytest CI]	2026 · GitHub
- Architected an event-driven streaming pipeline ingesting CARLA-style telemetry using Pub/Sub attributes and event-time semantics in Apache Beam. - Implemented stateful event_id deduplication, JSON schema validation, and automated Dead-Letter Queue (DLQ) routing for corrupted payloads. - Engineered partitioned and clustered BigQuery tables with mandatory partition filters to optimize analytical performance and cost. - Orchestrated daily data quality reporting with Apache Airflow in Docker Compose with automated verification in GitHub Actions CI.	
AutoSense AI: Predictive Vehicle Diagnostics & Agent [FastAPI, IsolationForest, RAG, Ollama LLM, MCP, React / Vite]	2026 · GitHub
- Engineered a pre-fault diagnostic system combining IsolationForest anomaly detection and vehicle baselines to detect degradation before DTCs. - Audited on 39,779 real OBD records; developed a 5-of-7 temporal alert stabilization filter reducing nuisance false alarms by 93.44%. - Integrated local TF-IDF RAG retrieval for OEM repair procedures and an autonomous agent with MCP tools and deterministic safety gates.	
CAN-Bus Diagnostic & Health Simulator [python-can, UDS (ISO 14229), DBC Network, Streamlit, SocketCAN]	2026 · GitHub
- Simulated multi-ECU CAN communication (Engine, Dynamics, Battery) using DBC definitions with modulo-16 alive counters and checksum validation. - Implemented UDS application services (0x10 Session Control, 0x22 Read Data By ID, 0x19 DTC Info) with automated YAML fault injection. - Maintained >70% branch test coverage with strict Ruff, Black, and MyPy quality gates executed in Linux/Windows GitHub Actions CI.	
Scenario-Centric AD Dataset & Trajectory Pipeline (Master Thesis) [CARLA 0.9.x, ADAS, Python, OpenDRIVE, Parquet]	TU Chemnitz · Current
- Generated highway merge, lane-change, and collision-avoidance scenarios in CARLA, extracting synchronized 3D actor trajectories and road topology. - Engineered an automated ingestion pipeline structuring raw simulation sensor frames into validated, columnar Parquet schemas for ML models.	

PROFESSIONAL EXPERIENCE

Working Student — AI Software Solutions DiPP GmbH	Jan 2025 – Jul 2025 Germany
- Integrated AI & RAG capabilities into CRM with FastAPI, Ollama LLMs, MCP tooling, and IONOS Cloud services. - Implemented document chunking, embeddings retrieval, and automated technical data workflows.	
Translator — German Patent & Technical Data Ascent Staffing (Client: Clarivate Analytics)	Dec 2021 – Jul 2022 India
Analyzed and translated German patent claims/specifications into English with strict technical terminology validation and QA.	
Associate Language Specialist — ML Data GlobalLogic Technologies	Oct 2020 – Dec 2021 India
Processed German retail datasets for NLP/ML pipelines; performed data annotation, classification, and model evaluation.	
Senior Process Executive (Client: Meta / Facebook) Cognizant Technology Solutions	Sep 2017 – May 2019 India
Automated Marketplace data workflows and human-in-the-loop evaluation; improved accuracy from ~64% to 76%.	
Practitioner (Client: Google) Concentrix Daksh Services	Mar 2016 – May 2017 India
Supported Google Ads automation via quality checks and error analysis; improved results from ~72% to 79%.	

EDUCATION

M.Sc. Automotive Software Engineering TU Chemnitz, Germany	Current 90/120 ECTS
Thesis: Scenario-Centric Datasets & Trajectories for AD in CARLA.	
M.Tech. Embedded Systems MLR Institute of Technology, India	2017 – 2019 First Class (CGPA 7.33)
B.Tech. Electrical & Electronics Eng. JNTU Hyderabad, India	2009 – 2013 First Class (62.67%)

CREDENTIALS

- Google Cloud Data Engineering Program
- Python Essentials 1 & 2 (Cisco / OpenEDG)
- SQL Certified (HackerRank)

LANGUAGES

- German: C1 (TU Chemnitz UNICert III / Goethe certified)
- English: C1 / Fluent (Master's conducted in English)
- Hindi: Fluent · Telugu: Native